

Eduardo Venegas

Data & AI Engineer

EDUCATION

MSc. Data Science and Artificial Intelligence Universität des Saarlandes 10/2024 - 09/2026 Saarbrücken, Germany

- Coursing: Statistics | ML Cybersecurity | NLP | Machine Translation | Image Processing & Computer Vision.
- Thesis (Seminar Grade 1.0):** Evaluating Multi-Turn Agentic RAG Systems (In collaboration with BMW & DFKI).

BSc. Computer Science Tecnológico de Monterrey GDL

08/2019 - 12/2023 Guadalajara, Mexico

- Grade: 94.6 / 100. Five times best software engineering GPA. **60% tuition** scholarship due to academic merit.
- ML | Time series | Algorithms & Optimization | Cloud Computing & Distributed | Vehicle User Interfaces | PLC.

Computer Science Exchange Student Universität Ulm

10/2022 - 09/2023 Ulm, Germany

- Full scholarship** recipient from DAAD. Computer Vision II | Autonomous Driving and Assistance Systems | Statistics.

Technologist Degree in Automation and Robotics CETI Colomos

08/2015 - 06/2019 Guadalajara, Mexico

- Control Laboratory (I & II) | Industrial Robotics | PLCs | Digital Electronics | Controllers Laboratory | Sensor Systems

SKILLS

Data

RAG LangChain LangGraph HuggingFace MCP R SQL TensorFlow Knowledge Graphs

Software

JavaScript TypeScript Python C C++ React Angular Flask Node Ionic/Tauri Matlab

Systems

Linux Docker Kubernetes AWS CI/CD Grafana MQTT PLC Arduino

WORK EXPERIENCE

Data & AI Engineer Masterarbeit BMW & Deutsches Forschungszentrum für KI (DFKI) 03/2026 - 09/2026 Munich, Germany

- Topic:** Evaluating Multi-Turn Agentic RAG Systems: A Synthetic Benchmark Generation Approach.
- Architected an evaluation framework for **multi-turn tool-using Agentic RAG** systems enabling precise failure localization.
- Built a **multi-turn synthetic benchmark pipeline** that combines deterministic **task planning** with diverse user **personas**.
- Engineered an **unsupervised retrieval engine** optimized to audit a **RAG system** across **4M unannotated** corporate records.

AI Research Engineer Werkstudent August-Wilhelm Scheer Institute

10/2024 - 02/2026 Saarbrücken, Germany

- Designed and deployed a **KG-RAG** over unstructured technical documentation using **LangChain** and **HuggingFace**.
- Developed an **Autoencoder model** for real-time **anomaly detection** across **distributed edge devices** via **Federated Learning**.

Big Data Platform Engineer Oracle

11/2023 - 10/2024 Guadalajara, Mexico

- Engineered **CI/CD testing pipelines** and deployment tools of **big data clusters** on **Oracle Cloud Infrastructure (OCI)**.
- Maintained automation frameworks within the **Big Data Service** team under **Oracle Analytics**.

Data Science & Analytics Intern BMW Group

03/2023 - 09/2023 Munich, Germany

- Developed a high-accuracy **anomaly detection algorithm** specialized in identifying **external battery replacements**.
- Engineered scalable **time-series data pipelines** using **PySpark** to process raw vehicle telemetry and cloud logs.
- Optimized data flows, resolving missing database values to increase data reliability and a **62% increase in service utilization**.
- Allowed adjusting predictive model parameters and receive **real-time system performance** outputs in under 8 seconds.

AI & DevOps Engineer Xira

01/2022 - 04/2022 Mexico City, Mexico

- Managed** over distributed server infrastructure, established service monitoring and **enhanced** engineering processes.
- Built an API management platform, integrating payments and subscription systems using **NextJS**, **Stripe** and **Docker**.

Production Engineer Fellow Major League Hacking & Meta

06/2021 - 09/2021 New York, USA

- Selected **internationally** among 120 out of 20,000 applicants. Implemented monitoring with **Prometheus** and **Grafana**.
- Automated **CI/CD** workflow for efficient deployment to a production server using **Docker** in an **AWS EC2** instance

Machine Learning & Full Stack Engineer Jiit Technologies

10/2020 - 06/2021 Guadalajara, Mexico

- Designed a custom sales forecasting and data analytics API pipeline using **Flask**, **SciPy** and **Prophet**.
- Built an **ERP** platform from scratch to handle multi-module operations (inventory, logistics, and finance) utilizing **Angular**.

PROJECTS

VectorDrive-Route: Autonomous BEV Trajectory Planning

PyTorch ResNet Computer Vision Github

- Built a **PyTorch** pipeline using **nuScenes**, **ResNet** and **depth lifting** to map multi-camera feeds into a **Bird's-Eye-View** grid.
- Designed a **temporal motion planning head** with **drivable-area** and **occupancy decoders** to generate safe paths.

Daten-Mapping-Framework für die AAS

AAS Industry 4.0 LLM

DOI: 10.17560/atp.v68i5.2836

- Developed a mapping **framework** from schemas to **Asset Administration Shell** to achieve interoperability in **Industry 4.0**.
- Designed a dual alignment strategy separating standard **relational** tables from **hierarchical** assets.

See Your World

Tensorflow Google Cloud Flask Github

- Crafted a hackathon **award-winning** app for the **visually impaired**, supporting **11 languages**.
- Engineered **Image Captioning** DL model with **TensorFlow**, deployed using **Flask**, **Docker**, and **Google Cloud VM**.

CERTIFICATIONS

- Mathematics for Data Science Specialization
- Data Science Professional Certificate
- Python Data Science Professional Certificate
- Deep Learning Professional Certificate

Higher School of Economics - 2021

Harvard - 2020

IBM - 2020

IBM - 2020

LANGUAGES

Spanish Native ●●●●●●

English C1 ●●●●●●

German C1 ●●●●●○

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